Use sub-query to solve following problems.

Note: To solve below queries, use “sales” database

1.

Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
Command Prompt - mysql -u

d6_96728_shaurya> select * from customers;
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2001 | Hoffman  | London   |    100 | 1001 |
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2006 | Clemens  | London   |    100 | 1001 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
| 2007 | Pereira  | Rome     |    100 | 1004 |
+------+----------+----------+--------+------+
7 rows in set (0.005 sec)

d6_96728_shaurya> select * from orders;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
10 rows in set (0.005 sec)

d6_96728_shaurya> select snum from customers where cname = "Cisneros";
+------+
| snum |
+------+
| 1007 |
+------+
1 row in set (0.005 sec)

d6_96728_shaurya> select * from orders where snum = (select snum from customers where cname = "Cisneros");
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.006 sec)
```

7.

Extract all the orders of Motika.

```
d6_96728_shaurya> select * from orders where snum = (select snum from salespeople where sname = "Motika")
    -> ;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.008 sec)

d6_96728_shaurya>
```

---

8. Find all the orders of salespeople servicing customers in London.

USE IN IF A SUBQUERY IS THROWING MORE THAN ONE OUTPUT

```
d6_96728_shaurya> select * from orders where snum IN (select snum from customers where city = "london");
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.028 sec)
```

---

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
d6_96728_shaurya> select cnum from customers where city = "London";
+------+
| cnum |
+------+
| 2001 |
| 2006 |
+------+
2 rows in set (0.007 sec)

d6_96728_shaurya> select MIN(amt) from orders where cnum IN (select cnum from customers where city = "London");
+----------+
| MIN(amt) |
+----------+
|   767.19 |
+----------+
1 row in set (0.008 sec)

d6_96728_shaurya> select * from orders where amt > (select MIN(amt) from orders where cnum IN (select cnum from customers where city = "London"));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.008 sec)
```

4.

Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
Command Prompt - mysql -u

d6_96728_shaurya> select snum from salespeople where sname = "Serres";
+------+
| snum |
+------+
| 1002 |
+------+
1 row in set (0.005 sec)

d6_96728_shaurya> select min(rating) from customers where snum = (select snum from
salespeople where sname = "Serres");
+-------------+
| min(rating) |
+-------------+
|         200 |
+-------------+
1 row in set (0.006 sec)

d6_96728_shaurya> select * from customers where rating >= (select min(rating) from
customers where snum = (select snum from salespeople where sname = "Serres"));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.007 sec)

d6_96728_shaurya>
```

11.

Select customers who have a greater rating than any other customer in Rome.

```
d6_96728_shaurya> select * from customers where rating >= (select max(rating) from
customers where city = "rome");
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.006 sec)
```

12.

Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’.

```
d6_96728_shaurya> select * from orders where amt > (select min(amt) from orders whe
re odate = "1990-10-04");
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.007 sec)
```

13.

Find all orders with amounts smaller than any amount for a customer in San Jose.

```
d6_96728_shaurya> select * from orders where amt < (select max(amt) from orders whe
re cnum in ( select cnum from customers where city = "San Jose"));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.008 sec)
```

2.

Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
d6_96728_shaurya> select o.amt , c.cname, c.rating from customers c
    -> left join orders o ON o.cnum = c.cnum
    -> where amt > (select avg(amt) from orders);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.007 sec)

d6_96728_shaurya>
```

14.

Select those customers whose rating are higher than every customer in London.

```
d6_96728_shaurya> select * from customers where rating > (select max(rating)
 from customers where city = "London");
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.006 sec)
```

5.

Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
d6_96728_shaurya> select * from salespeople where snum not in (select SSNUM
from (select s.snum as ssnum, s.sname, s.city AS scity, c.cname, c.city as c
city, c.snum as csnum from salespeople s right join customers c on c.snum =
s.snum) as t where ccity = scity);
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.008 sec)

d6_96728_shaurya>
```

STEP 1:

```
d6_96728_shaurya> select s.snum as ssnum, s.sname, s.city AS scity, c.cname,
 c.city as ccity, c.snum as csnum from salespeople s right join customers c
on c.snum = s.snum;
+-------+---------+-----------+----------+----------+-------+
| ssnum | sname   | scity     | cname    | ccity    | csnum |
+-------+---------+-----------+----------+----------+-------+
|  1001 | Peel    | London    | Hoffman  | London   |  1001 |
|  1003 | Axelrod | New York  | Giovanni | Rome     |  1003 |
|  1002 | Serres  | San Jose  | Liu      | San Jose |  1002 |
|  1002 | Serres  | San Jose  | Grass    | Berlin   |  1002 |
|  1001 | Peel    | London    | Clemens  | London   |  1001 |
|  1007 | Rifkin  | Barcelona | Cisneros | San Jose |  1007 |
|  1004 | Motika  | London    | Pereira  | Rome     |  1004 |
+-------+---------+-----------+----------+----------+-------+
7 rows in set (0.006 sec)
```

STEP 2:

```
d6_96728_shaurya> select * from (select s.snum as ssnum, s.sname, s.city AS
scity, c.cname, c.city as ccity, c.snum as csnum from salespeople s right jo
in customers c on c.snum = s.snum) as t where ccity = scity;
+-------+--------+----------+---------+----------+-------+
| ssnum | sname  | scity    | cname   | ccity    | csnum |
+-------+--------+----------+---------+----------+-------+
|  1001 | Peel   | London   | Hoffman | London   |  1001 |
|  1002 | Serres | San Jose | Liu     | San Jose |  1002 |
|  1001 | Peel   | London   | Clemens | London   |  1001 |
+-------+--------+----------+---------+----------+-------+
3 rows in set (0.008 sec)
```

STEP 3:

```
d6_96728_shaurya> select SSNUM from (select s.snum as ssnum, s.sname, s.city
 AS scity, c.cname, c.city as ccity, c.snum as csnum from salespeople s righ
t join customers c on c.snum = s.snum) as t where ccity = scity;
+-------+
| ssnum |
+-------+
|  1001 |
|  1002 |
|  1001 |
+-------+
3 rows in set (0.006 sec)
```

STEP 4:

```
d6_96728_shaurya> select * from salespeople where snum not in (select SSNUM
from (select s.snum as ssnum, s.sname, s.city AS scity, c.cname, c.city as c
city, c.snum as csnum from salespeople s right join customers c on c.snum =
s.snum) as t where ccity = scity);
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.008 sec)
```

STEP 5:

```
d6_96728_shaurya> select sname, snum, city from salespeople where snum not i
n (select SSNUM from (select s.snum as ssnum, s.sname, s.city AS scity, c.cn
ame, c.city as ccity, c.snum as csnum from salespeople s right join customer
s c on c.snum = s.snum) as t where ccity = scity);
+---------+------+-----------+
| sname   | snum | city      |
+---------+------+-----------+
| Motika  | 1004 | London    |
| Rifkin  | 1007 | Barcelona |
| Axelrod | 1003 | New York  |
+---------+------+-----------+
3 rows in set (0.010 sec)
```

9.

Find names and numbers of all salesperson who have more than one customer.

```
d6_96728_shaurya> select sname, snum from salespeople where snum in (select
snum from customers GROUP BY snum HAVING count(snum) > 1);
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.007 sec)
```

10.

Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
d6_96728_shaurya> select Q1.snum, sname, city, customer_count from (select *
 from salespeople where snum in (select snum from customers GROUP BY snum HA
VING count(snum) > 1)) as Q1 Left join (select snum, count(snum) AS customer
_count from customers GROUP BY snum HAVING count(snum) > 1) Q2 ON Q1.snum =
Q2.snum;
+------+--------+----------+----------------+
| snum | sname  | city     | customer_count |
+------+--------+----------+----------------+
| 1001 | Peel   | London   |              2 |
| 1002 | Serres | San Jose |              2 |
+------+--------+----------+----------------+
2 rows in set (0.008 sec)

d6_96728_shaurya>
```

Step 1 :

```
d6_96728_shaurya> select snum, count(snum) AS customer_count from customers
GROUP BY snum;
+------+----------------+
| snum | customer_count |
+------+----------------+
| 1001 |              2 |
| 1003 |              1 |
| 1002 |              2 |
| 1007 |              1 |
| 1004 |              1 |
+------+----------------+
5 rows in set (0.005 sec)
```

Step 2:

```
d6_96728_shaurya> select snum, count(snum) AS customer_count from customers
GROUP BY snum HAVING count(snum) > 1;
+------+----------------+
| snum | customer_count |
+------+----------------+
| 1001 |              2 |
| 1002 |              2 |
+------+----------------+
2 rows in set (0.009 sec)
```

Step 3:

```
d6_96728_shaurya> select * from (select * from salespeople where snum in (se
lect snum from customers GROUP BY snum HAVING count(snum) > 1)) as Q1;
+------+--------+----------+------+
| snum | sname  | city     | comm |
+------+--------+----------+------+
| 1001 | Peel   | London   | 0.12 |
| 1002 | Serres | San Jose | 0.13 |
+------+--------+----------+------+
2 rows in set (0.010 sec)
```

Step 4:

```
d6_96728_shaurya> select * from (select * from salespeople where snum in (se
lect snum from customers GROUP BY snum HAVING count(snum) > 1)) as Q1 Left j
oin (select snum, count(snum) AS customer_count from customers GROUP BY snum
 HAVING count(snum) > 1) Q2 ON Q1.snum = Q2.snum;
+------+--------+----------+------+------+----------------+
| snum | sname  | city     | comm | snum | customer_count |
+------+--------+----------+------+------+----------------+
| 1001 | Peel   | London   | 0.12 | 1001 |              2 |
| 1002 | Serres | San Jose | 0.13 | 1002 |              2 |
+------+--------+----------+------+------+----------------+
2 rows in set (0.009 sec)
```

Step 5:

```
d6_96728_shaurya> select Q1.snum, sname, city, customer_count from (select *
 from salespeople where snum in (select snum from customers GROUP BY snum HA
VING count(snum) > 1)) as Q1 Left join (select snum, count(snum) AS customer
_count from customers GROUP BY snum HAVING count(snum) > 1) Q2 ON Q1.snum =
Q2.snum;
+------+--------+----------+----------------+
| snum | sname  | city     | customer_count |
+------+--------+----------+----------------+
| 1001 | Peel   | London   |              2 |
| 1002 | Serres | San Jose |              2 |
+------+--------+----------+----------------+
2 rows in set (0.009 sec)
```

3.

Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
d6_96728_shaurya> SELECT snum, SUM(amt) AS total_sales
    -> FROM orders
    -> GROUP BY snum
    -> HAVING SUM(amt) > (SELECT MAX(amt) FROM orders);
+------+-------------+
| snum | total_sales |
+------+-------------+
| 1001 |    15382.07 |
+------+-------------+
1 row in set (0.081 sec)
```